# AASTHA THAKUR

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## Education

**Jaypee University Of Information Technology** **2022 - 2026**
*Bachelor of Technology in Computer Science* *CGPA: 7.39*

**ST Mary's School, Sundernagar** **2021 - 2022**
*Higher Secondary* *Percentage: 77.6%*

**ST Mary's School, Sundernagar** **2020-2021**
*Secondary* *Percentage: 75%*

## Relevant Coursework

- DBMS
- Data Structures and Algorithms
- Machine Learning
- Data Science
- Probability and Stats
- Discrete Computational Mathematics
- Operating System
- DAA

## Skills

**Languages**: Python, SQL, C, C++, Javascript
**Technologies/Frameworks**:MySQL, AI/ML, Pandas, Streamlit, Numpy, UiPath, SAP, Worksoft, RAG
**Developer Tools**: Git, Github, Google Collab, Anaconda, Jupyter Notebook,VS Code
**Soft Skills**: Problem Solving, Self-learning, Adaptability, Team Collaboration

## Projects/Experience

**Associate Test Engineer** | *TestingXperts* **Jan 2026 – Present**

- Performed **manual and automation testing** to validate application functionality and identify defects.
- Worked with **UiPath, Worksoft, SAP, and Selenium** for automation testing activities.
- Automated test scenarios for an **e-commerce website** to validate user workflows.
- Contributed to testing activities for the **Air Canada and CorpCu** projects.
- Performed **migration testing**, including test case preparation, execution, and defect reporting.
- Currently supporting test case preparation and execution for an internal project.

**Wheelchair Object Detection & Gesture Control** | *Python, OpenCV, YOLOv5, MediaPipe, ROS* - July 2025

- Completed a Computer Vision Research Internship at IIT Mandi, curating and collecting a custom dataset for 5 distinct hand gestures.
- Trained a robust gesture recognition model utilizing Gaussian Mixture Models (GMM) for background subtraction and feature extraction.
- Integrated OpenCV, MediaPipe, and YOLOv5 for precise real-time object detection and hand gesture tracking.
- Connected the trained vision model with the Robot Operating System (ROS) to enable seamless assistive wheelchair navigation controlled via hand gestures.
- Evaluated system performance under varying environmental and lighting conditions, ensuring high reliability and low latency for real-time mobility assistance.

**PDF RAG Chatbot** | *Python, LangChain, Google Gemini, FAISS, Streamlit* | GitHub **May 2026**

- Developed a production-ready **Retrieval Augmented Generation (RAG) chatbot** leveraging LangChain and Google Gemini to answer queries from multiple PDF documents.
- Implemented text extraction and recursive chunking with **Sentence Transformers**, storing embeddings in a **FAISS** vector database for semantic search.
- Integrated LangChain's **ConversationBufferMemory** to support context retention and multi-turn chat interactions.
- Designed a clean **Streamlit** web interface featuring dynamic environment configuration, robust error handling, and chat history management.
- Optimized retrieval accuracy by tuning chunk size and overlap parameters, reducing LLM hallucination and improving response relevance for complex queries.